

Learn Cryptocurrency: Beginner's Guide

Lesson 3: Key Cryptocurrency Terms

Key Cryptocurrency Terms - Study Notes

Key Concepts

1. **Wallets** – Software or hardware tools that store your crypto /keys and let you send/receive assets.
2. **Private & Public Keys** – Cryptographic pair where the private key signs transactions and the public key (or its address) receives funds.
3. **Mining** – The process of validating transactions and adding new blocks to a blockchain, typically rewarded with new coins.
4. **Gas** – The fee paid to execute operations on a blockchain (most commonly Ethereum).
5. **Token** – A digital asset built on top of an existing blockchain (e.g., ERC 20 tokens on Ethereum).

Summary

Cryptocurrency basics revolve around the concepts of **keys** and **wallets**. A wallet does not actually hold coins; it holds the private keys that give you control over the coins associated with the corresponding public addresses. The **private key** must stay secret because anyone who knows it can move the funds, while the **public key** (or derived address) can be shared freely to receive payments.

Mining is the engine that secures many blockchains. Miners solve computational puzzles, validate transactions, and bundle them into blocks. In proof of work (PoW) systems like Bitcoin, miners are rewarded with newly minted coins and transaction fees. In proof of stake (PoS) systems, validators are chosen based on the amount of stake they hold, but the term “mining” is still often used colloquially.

On programmable blockchains such as Ethereum, **gas** measures the computational effort required to run operations (e.g., smart contract calls). Users pay gas in the native token (ETH) to compensate validators for processing their transactions. Finally, a **token** is any digital asset that lives on top of an existing blockchain. Tokens can represent utility, governance rights, stablecoins, or even real world assets, and they inherit the security of the underlying chain.

Important Points

- **Wallet Types**:
 - **Hot wallets** (online, convenient, higher risk).
 - **Cold wallets** (offline hardware or paper, more secure).
- **Key Management**:
 - Never share your private key or seed phrase.
 - Back up the seed phrase in a safe, offline location.
- **Public Key vs. Address**:
 - Public key != hashed != encoded != address (e.g., Bitcoin's Base58Check, Ethereum's hex address).
- **Mining Rewards**:

- Block reward + transaction fees (or “gas fees”).
- Reward halves periodically in Bitcoin (halving events).
 - **Gas Mechanics (Ethereum example)**:
- **Gas limit** = max units you're willing to spend.
- **Gas price** = amount of ETH per gas unit (gwei).
- Total fee = gas limit × gas price.
 - **Token Standards**:
- ERC 20 (fungible), ERC 721 (non fungible), ERC 1155 (multi token).

Examples

1. **Sending Bitcoin from a Mobile Wallet**

- Alice opens her mobile wallet (a hot wallet) and enters Bob's Bitcoin address (derived from his public key).
- The wallet creates a transaction, signs it with Alice's private key, and broadcasts it.
- Miners include the transaction in a block; Alice pays a small transaction fee (in satoshis) to the miner.

2. **Interacting with an ERC 20 Token on Ethereum**

- Bob wants to swap 100 USDT (an ERC 20 stablecoin) for ETH on a decentralized exchange.
- He initiates the swap in his wallet; the wallet estimates the required gas (e.g., 120,000 gas).
- Bob sets a gas price of 30 /gwei; total fee = 120,000 /× /30 /gwei "H 0.0036 /ETH.
- The transaction is signed with Bob's private key, sent to the network, and validators execute the smart contract code, moving USDT and ETH accordingly.

Quick Review

1. **What is the role of a private key in a cryptocurrency wallet?**
2. **How does gas differ from a transaction fee on Bitcoin?**
3. **Name two main types of wallets and a key advantage of each.**
4. **What is a token, and how does it relate to the underlying blockchain?**
5. **Why do miners receive both a block reward and transaction fees?**

Further Reading

- **Public Key Cryptography Basics** – deeper dive into elliptic curve cryptography used by most blockchains.
- **Proof of Work vs. Proof of Stake** – consensus mechanisms and their impact on mining/validation.
- **Ethereum Gas Optimization** – strategies for reducing gas costs when interacting with smart contracts.
- **Token Standards Overview** – ERC 20, ERC 721, ERC 1155, BEP 20, etc.
- **Secure Wallet Practices** – hardware wallets, seed phrase management, and phishing avoidance.

